Panlin Li

High-Dimensional Statistical Learning · Functional Data Analysis



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Education

Xinjiang Agricultural University · B.S. in Mathematics & Applied Mathematics Sep. 2023 – Jul. 2027

GPA 3.88/5.0 · **Coursework Top 1.5%** · **University-wide Rank 1st**

TOEFL 110 · Advisor: Prof. Hua Huang

12	9	1	15	13	4
Papers	First-author	In submission (SCI Q1)	Projects (2 nat'l)	Awards (5 nat'l)	Software copyrights

Honors & affiliation: National Scholarship; 9 volunteer service certificates & an Outstanding Team award, CYL Central Committee Youth Volunteer Action Guidance Center / Chinese Young Volunteers Association Secretariat; Outstanding Student Pacesetter, Outstanding Student and Innovation & Entrepreneurship Star (3 consecutive years each), Science & Innovation Pioneer Scholarship, Outstanding Class Collective (as class monitor) — XJAU; Member, China Society for Industrial and Applied Mathematics.

Research & Projects

Xinjiang NSF General Program · Participant — Spatial–Spectral Fusion for Functional Modeling — 2026 – Present

- **Situation** Hyperspectral imaging carries image and spectrum together, yet conventional pipelines process the two separately and concatenate them — breaking spatial structure while still facing band redundancy and the curse of dimensionality.
- **Action** I own image acquisition and preprocessing (9-member team, 10 months a year): ROI extraction, spectral correction, outlier removal and spline interpolation onto a common wavelength grid; then basis expansion turning discrete spectra into functional curves, with a basis-family and basis-number selection strategy settled by comparison.
- **Result** The pipeline now carries both representation routes — functional PCA and interval generalized PLS; the proposal further **seeded 3 student innovation projects** (1 regional, 2 university-level), whose sub-topics I lead-authored.

Provincial Project · Ongoing · PI — Shenji Miansuan — Sky-Ground Integrated Digital Cotton Field — 2026 – Present

- **Situation** Pest scouting, water-and-fertiliser decisions and residual-film management on large cotton farms still rely on manual inspection and experience, with scattered sensing hardware and no digital layer turning data into decisions.
- **Action** **Project PI**, leading a four-person team and an industry partner; built a sky-ground monitoring network from ESP32 nine-axis environmental sensing and drone multispectral imaging, an improved YOLOv5 pest recogniser and a water–nitrogen–carbon–film coupled irrigation and fertilisation model, with edge-cloud collaboration and federated learning keeping each farm's data on site.
- **Result** Entered the industry track (growth group) of the 12th Xinjiang Innovation & Entrepreneurship Competition with a partner company; the due-diligence review **recommended advancement**.

National Project · Completed · PI — Small-Sample Time-Series Forecasting for Energy Economics — 2024 – 2025

- **Situation** Energy-consumption series offer only a handful of annual points, where conventional forecasters overfit; grey models are viable here, but whether an integer accumulation order is optimal had never been tested.
- **Action** Lead-authored the design; re-engineered the kernel-regularized non-homogeneous grey model with conformable fractional accumulation (CFAKRNGM) and searched the continuous order by the Hiking Optimization Algorithm against a KRNGM baseline.
- **Result** MAPE fell from 0.0557 to 0.0522, but the optimum landed at $\alpha = 0.9998 / 0.9995$ — back at integer order, so **the fractional mechanism was never activated by the data**; the negative result is reported in full (co-first author).

Other 12 student innovation projects: National (1) — apple origin traceability (core member); **Provincial (4)** — Yunwen Jiangrong Hub (PI · Excellent), Apple SSC via FDA + image fusion (Excellent), Tongxi Zhixing visually-impaired navigation, hyperspectral multi-task detection; **University (7)** — nonlinear wave solutions of high-dimensional PDEs (PI · Excellent), cross-device spectral transfer for online rapid detection (PI), Zhitong smart cane (Excellent), Tianshan Pastoral Song (Excellent), non-traveling-wave solutions and localized excitations, AI pest & disease diagnosis, fruit & vegetable storage-quality forecasting; one of them recognized by the AI Ecosystem Group, Huawei Hangzhou Research Institute.

Publications

- **Li P.L.**, Li Y.C., Li L.J., et al. *Calibration transfer under limited target labels: a five-benchmark comparison of correction and representation methods*. Chemometrics and Intelligent Laboratory Systems. (**SCI Q1**; IF 3.8; Major Revision)
- **Li P.L.** (presenter), Yang S.Z., Huang H.* *Upper bound of reference-value reliability and a sampling design table for near-infrared calibration of apple SSC*. The 11th National Conference on Near-Infrared Spectroscopy, China.
- **Li P.L.**, Zhang N. *Optimization of NIPT Testing Timing and Fetal Anomaly Determination Models*. International Journal of Public Health and Medical Research, 2026, 6(2): 1–8.

- **Li P.L.**, Bao F.C., Zhang J.Y., et al. *Periodic Solutions to a New Class of KdV Equations*. Advances in Applied Mathematics, 2025, 14(5): 23–28.
- **Li P.L.**, Zhang N. *Prediction of optimal NIPT timing and analysis of influencing factors of fetal Y chromosome concentration based on robust Bayesian ridge regression*. Academic Journal of Computing & Information Science, 2025, 8(10): 99–107.
- **Li P.L.**, Zhang N. *Study on the determination of fetal chromosomal abnormalities based on multi-model regression*. Academic Journal of Mathematical Sciences, 2025, 6(2): 121–129.
- **Li P.L.**, Zhang N. *Study on the effect of n-hexane insolubles on pyrolysis product yields based on statistical regression analysis*. Highlights in Science, Engineering and Technology, 2024, 116: 270–276.
- **Li P.L.**, Zhang N. *Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis*. In: 2025 IEEE 3rd International Conference on Control, Electronics and Computer Technology (ICCECT), Jilin, China, 2025, pp. 998–1004.
- **Li P.L.**, Li J., Kadir A., et al. *Cultural Heritage and Rural Revitalization: Exploration and Reflections on China's Practice*. Smart Agriculture Guide, 2024, 4(21): 185–188.
- Zhang N., **Li P.L.**, Bao F.C., Liu X.Y. *Solitary Wave Solutions and Interaction Solutions of an Extended (3+1)-Dimensional Shallow Water Wave Equation*. Journal of Changchun Normal University, 2026, 45(4): 6–11.
- Jiao Q.T., **Li P.L.**, Mei T., et al. *ESP32-Based Multi-Sensor Near- & Far-Range Water Quality Monitoring System*. Internet of Things Technologies, 2026, 16(2): 9–12, 16.
- **Software Copyrights (4, as first holder):** Intelligent White Cane Vision Recognition System · Smart Cane Data Collection & Monitoring System · User Behavior Analysis and Recommendation System · Big-Data-Driven Market Analysis and Forecasting System.

Competitions & Awards

China International College Students' Innovation Competition (2024)	**National · Bronze** · Participant
11th National Undergraduate Statistical Modeling Contest	**National · Third Prize** · Team Lead
"Chia Tai Cup" National Market Survey & Analysis Contest (15th & 16th)	**National · Third Prize × 2** · Team Lead
"DataBee Cup" National Interview Survey Contest	**National · Third Prize** · Team Lead
China International College Students' Innovation Competition, Xinjiang Division (2025)	**Provincial · Bronze** · Team Lead
National Mathematical Modeling Contest, Xinjiang Division (2024)	**Provincial · Second Prize** · Team Lead
10th National Undergraduate Statistical Modeling Contest, Xinjiang Division	**Provincial · Second Prize** · Team Lead
14th China Innovation & Entrepreneurship Competition, Xinjiang Division	**Provincial · Third Prize** · Team Lead
Xinjiang Division Information Literacy Contest (2024)	**Provincial · Merit Award** · Team Lead

Student Experience

Class Monitor, Mathematics 2302, School of Mathematics & Physics, Xinjiang Agricultural University — Sep. 2023 – Jul. 2027
Class Monitor, 19th Youth Marxist Training Program, Xinjiang Agricultural University — May 2025 – May 2026
Head, Volunteer Association, School of Mathematics & Physics — Dec. 2024 – Dec. 2025

- Founded the 20-member "Shield of the Future" volunteer team; completed 9 volunteer service programs certified by the CYL Central Committee Youth Volunteer Action Guidance Center and the Chinese Young Volunteers Association Secretariat; the work was covered by *China Daily*, *People's Daily* and other outlets.